Method	Type	Bone marrow	Breast	Cervix	Colon	Lung	Avg
Self-Refine† (3 rounds)	Multi-turn	0.46	0.53	0.53	0.48	0.49	0.50
Multi-Agent Debate†	Multi-agent	0.54	0.51	0.53	0.54	0.51	0.52
MedAgents†	Domain multi-agent	0.47	0.50	0.62	0.45	0.48	0.50
AIAgents4Pharma†	KG-augmented	0.46	0.50	0.51	0.50	0.50	0.49
PBIO-Agent (Ours)	**Domain multi-agent**	**0.52**	**0.69**	**0.64**	**0.58**	**0.68**	**0.62**

Table 1: **Comparison with multi-agent/multi-turn designs on LINCSQA.** Binary AUROC across 5 tissue types. All methods use the same base LLM (DeepSeek-R1-Distill-Llama-8B). †Results reported from a single seed.

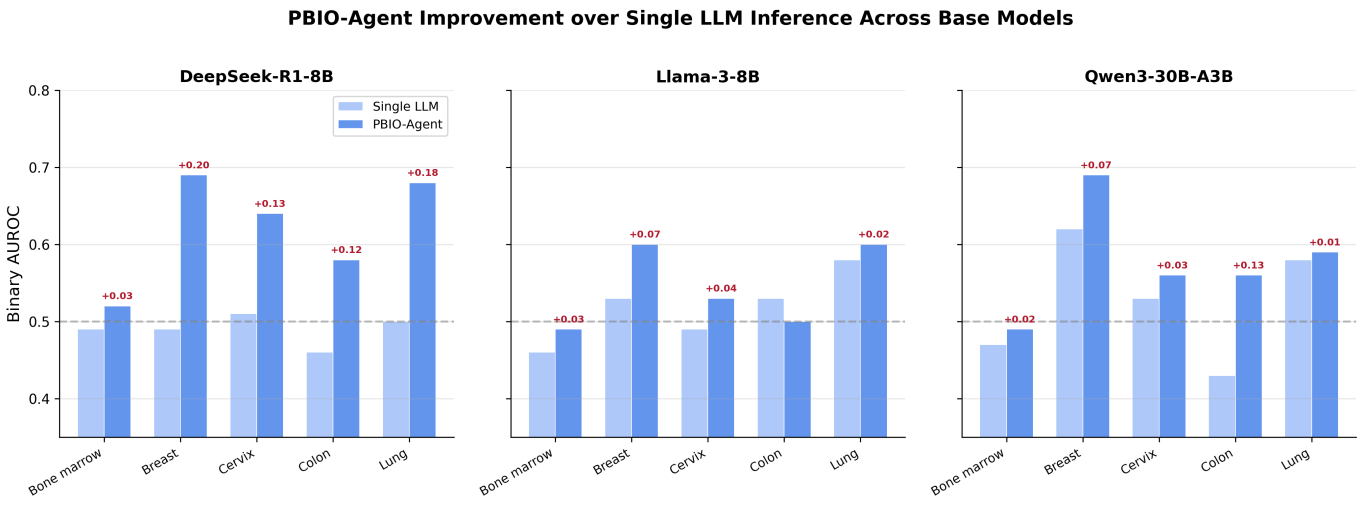

Figure 1: **PBIO-Agent with different base LLMs on LINCSQA.** Binary AUROC scores comparing single LLM inference vs. PBIO-Agent framework. The agent framework consistently improves over single LLM inference across all tested base models, confirming that performance gains stem from the agent design rather than the choice of base model.

Variant	Bone marrow	Breast	Cervix	Colon	Lung	Avg
Full PBIO-Agent	**0.52**	**0.69**	**0.64**	**0.58**	**0.68**	**0.62**
Single Agent†	0.51	0.68	0.58	0.57	0.65	0.60
w/o Context†	0.48	0.68	0.64	0.53	0.61	0.59
w/o Mechanism†	0.46	0.65	0.56	0.51	0.63	0.56
w/o Network†	0.49	0.68	0.57	0.56	0.63	0.59

Table 2: **Ablation on scientist agents.** Binary AUROC across 5 tissue types. "Full" = all 3 scientists + integration. "Single Agent" = integration agent only. The Mechanism Agent has the largest single-agent impact ($\Delta = -0.06$ avg), followed by Context Agent ($\Delta = -0.04$) and Network Agent ($\Delta = -0.04$).†Results reported from a single seed.

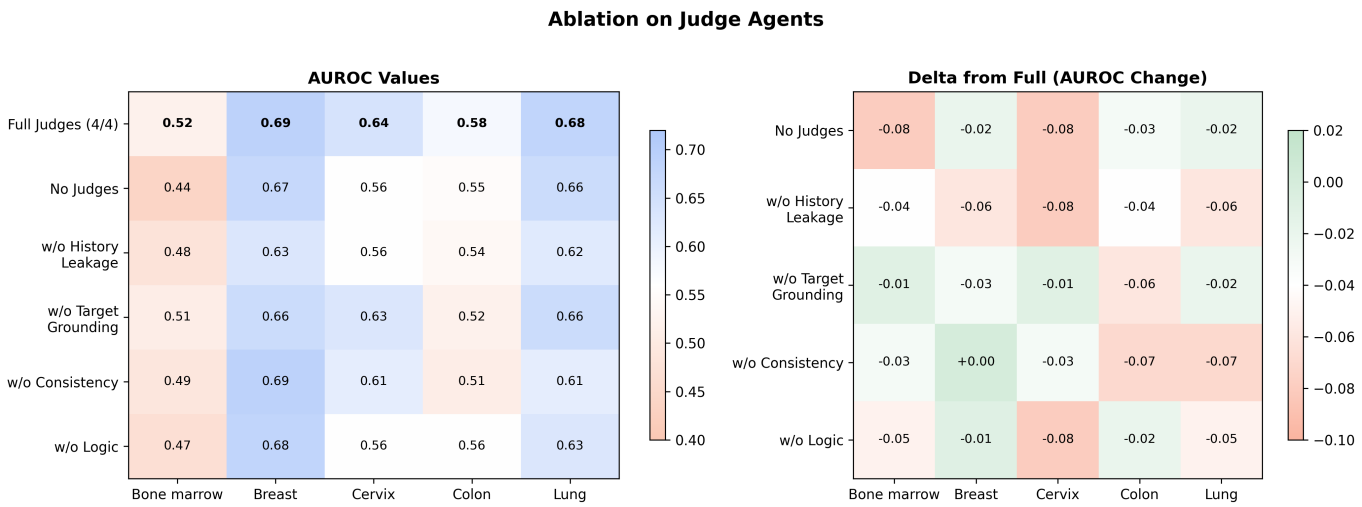

Figure 2: **Ablation on judge agents.** Left: AUROC values for each variant across tissues. Right: AUROC change (Δ) relative to the full system. History Leakage Judge is the most impactful individual judge (Δ=-0.06 avg), confirming that preventing ground-truth leakage is essential for prediction integrity.

Variant	Bone marrow	Breast	Cervix	Colon	Lung
Full PBIO-Agent	**0.52**	**0.69**	**0.64**	**0.58**	**0.68**
Single Agent† + No Judge	0.51	0.68	0.57	0.55	0.65
Multi-Agent† + No Judge	0.44	0.67	0.56	0.55	0.66
Single Agent† + Judge	0.51	0.68	0.58	0.57	0.65

Table 3: **Cross-component interaction ablation.** Isolating interactions between multi-agent and judge components. †Results reported from a single seed.

Metric	Value
Avg. input tokens per sample	19,685
Avg. output tokens per sample	1,878
Avg. total tokens per sample	21,564
Avg. LLM calls per sample	14.1
Avg. wall-clock time per sample	131.9s
Avg. retry count	0.76
Samples reaching max retry (%)	22.6%
Cost multiplier vs. single LLM	4.8×

Table 4: **Computational cost analysis of PBIO-Agent.**

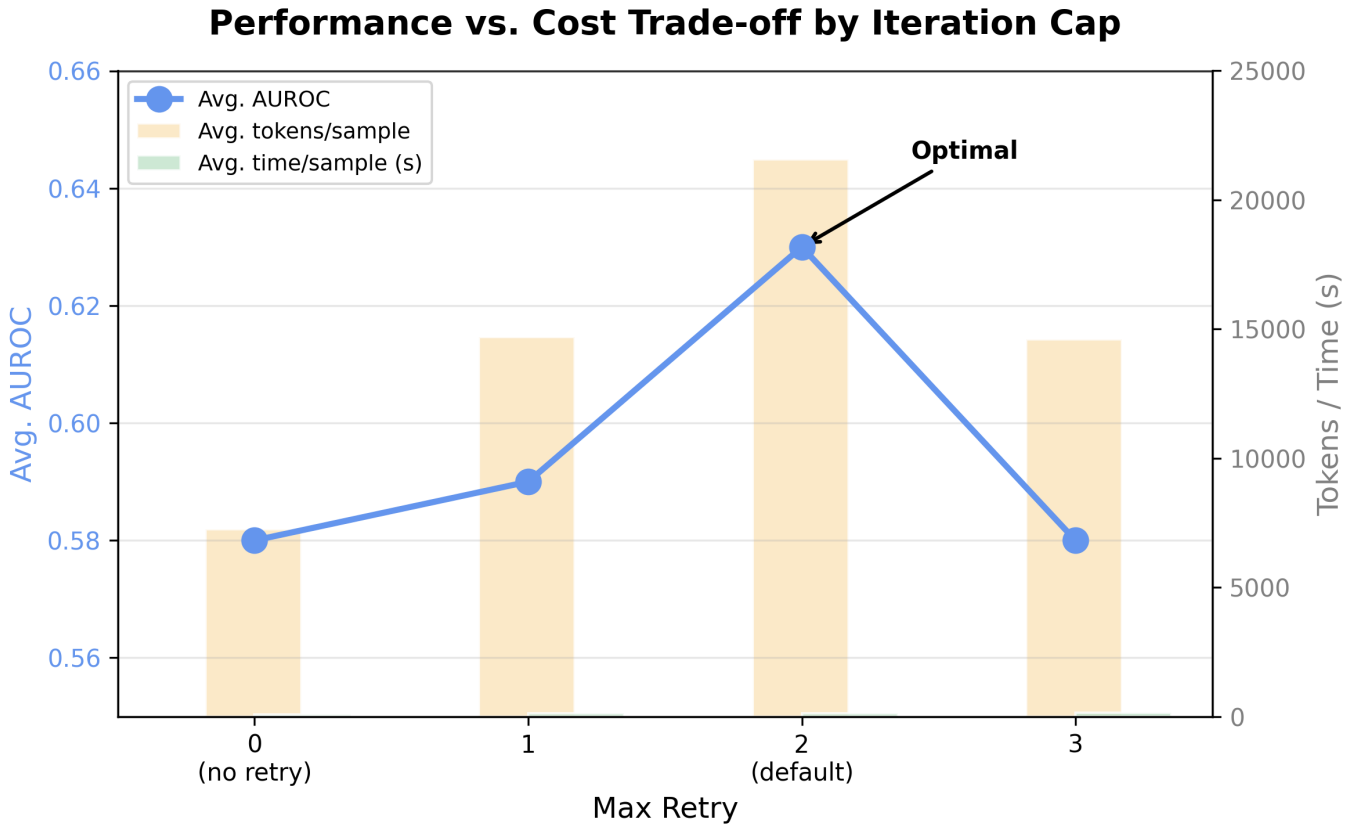

Figure 3: **Performance vs. cost trade-off by iteration cap.** AUROC (line) against token usage and wall-clock time (bars). max_retry=2 achieves the optimal accuracy–cost trade-off (+0.05 AUROC over no retry at 3.0× token cost). Increasing to max_retry=3 degrades AUROC back to 0.58 despite higher wall-clock time.

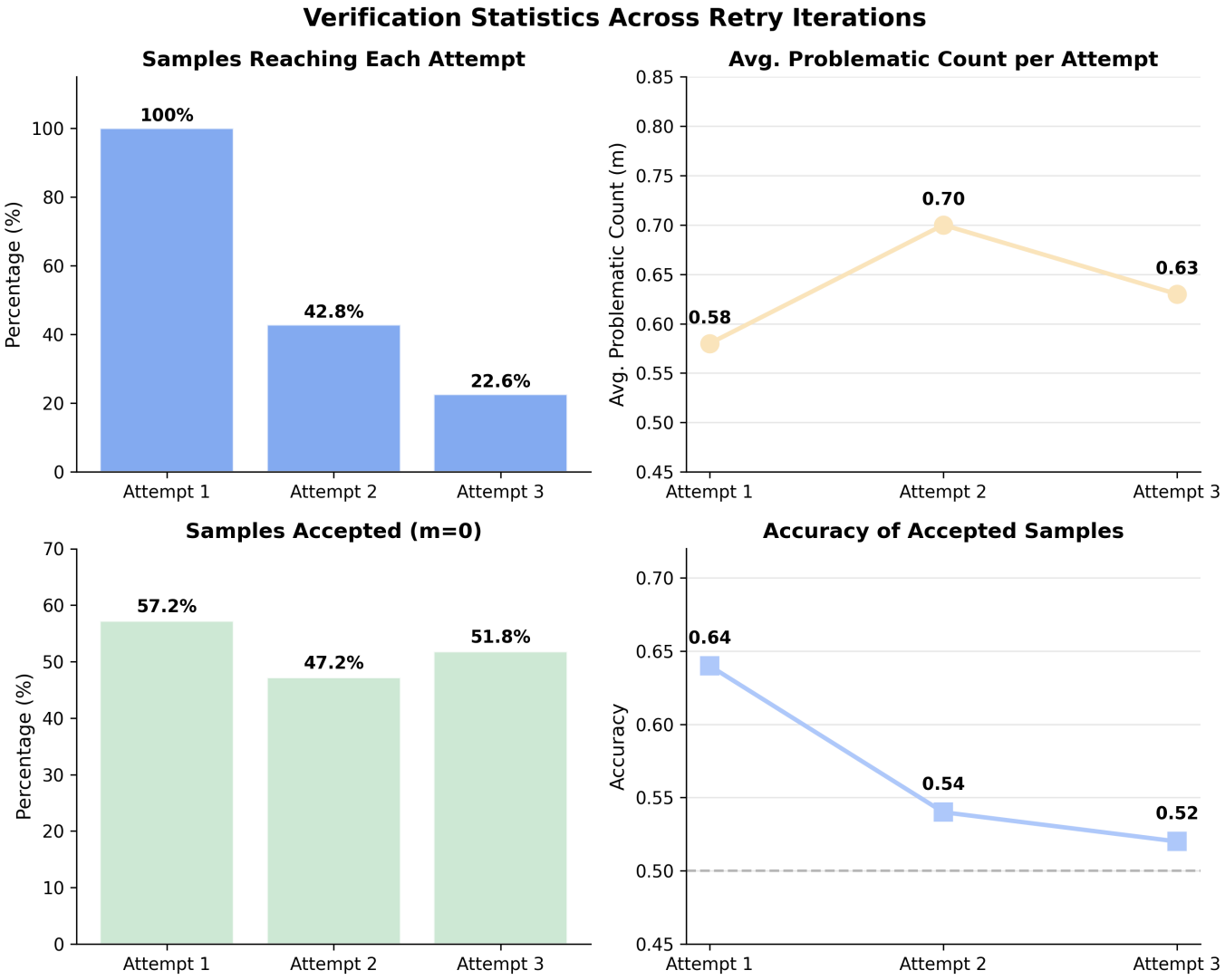

Figure 4: **Verification statistics across retry iterations.** (Top-left) Percentage of samples reaching each attempt. (Top-right) Average problematic count m does not monotonically decrease. (Bottom-left) Acceptance rate per attempt. (Bottom-right) Accuracy of accepted samples decreases across attempts, revealing natural difficulty stratification.

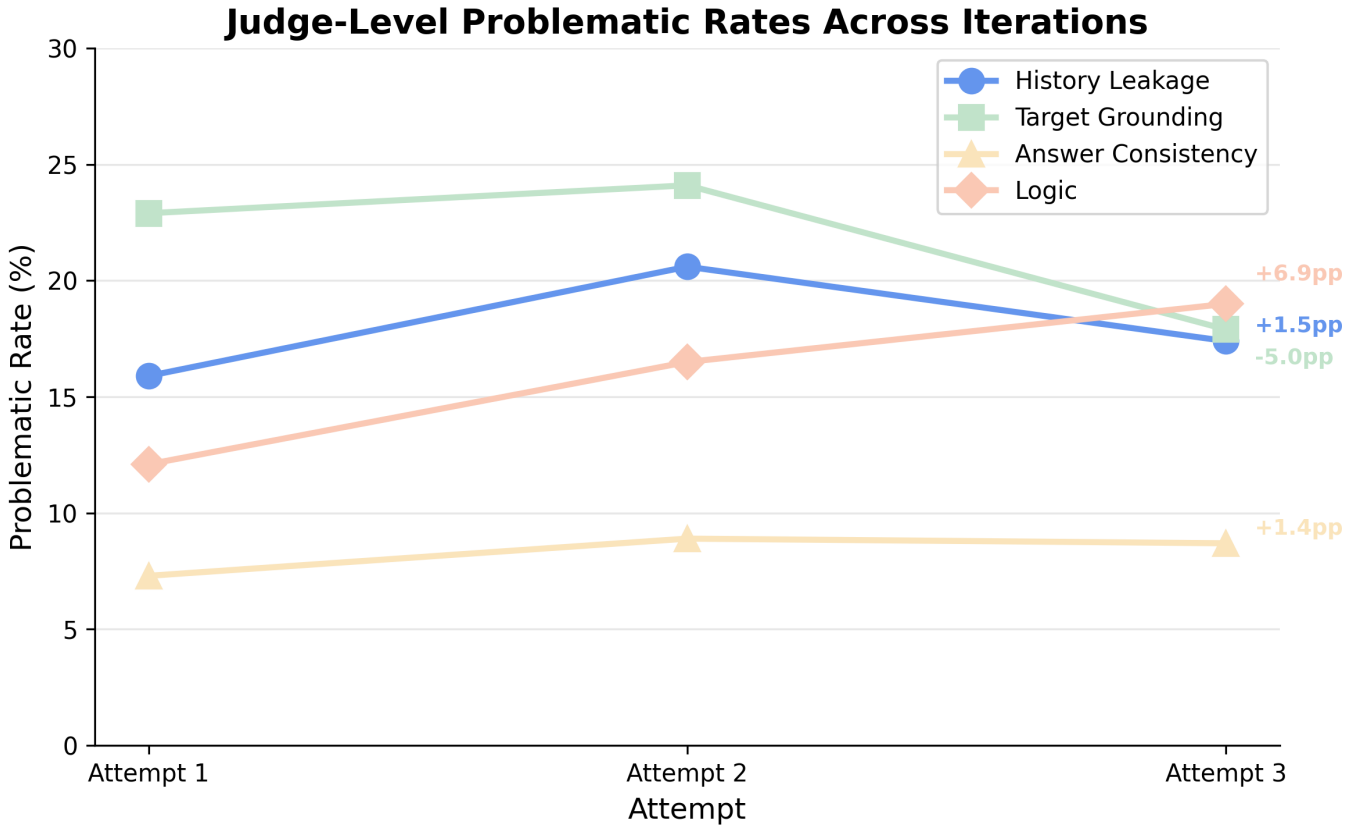

Figure 5: **Judge-level problematic rates across iterations.** Target Grounding improves with retries (22.9%→17.9%), while Logic problematic rates increase (12.1%→19.0%), indicating that repeated reasoning on difficult samples may introduce logical inconsistencies.

Model	Bone marrow	Breast	Cervix	Colon	Lung
PBIO-Agent (easy→hard)	0.52±.02	**0.69±.01**	**0.64±.02**	**0.58±.09**	**0.68±.03**
Random	**0.54±.04**	0.66±.03	0.58±.02	**0.58±.07**	0.63±.02
Reverse (hard→easy)	0.52±.01	0.65±.01	0.56±.02	0.56±.08	0.59±.01

Table 5: **Ablation on difficulty-aware ordering in progressive reasoning.** Binary AUROC scores across 5 tissue types. Mean ± std over 3 runs. Best per tissue in **bold**.